

2016.0 RANGE ROVER (LG), 501-20

SEATBELT SYSTEM

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the seatbelt system and operation, refer to the relevant description and operation section of the workshop manual.

SAFETY INFORMATION

WARNINGS:

- To avoid accidental deployment the back-up power supply must be depleted before beginning any work on the SRS system or its components. Failure to follow this instruction may result in personal injury
- Do not use a multimeter to probe an SRS module. It is possible for the power from the multimeter battery to trigger the activation of the module. Failure to follow this instruction may result in personal injury

NOTE:

Do not to use a cellular phone or to have a cellular phone in close proximity when working on the SRS system or components

Power supply depletion

Before beginning any work on the SRS system or related components:

1. Remove the ignition key
2. Disconnect the battery leads, ground first
3. Wait 2 minutes for the power circuit to discharge

There are comprehensive instructions on the correct procedures for SRS system repairs, refer to the relevant section of the workshop manual

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle

NOTE:

Check and rectify basic faults before beginning diagnostic routines including pinpoint tests

1. Verify the customer concern by operating the seatbelt
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Check for the installation of non-standard accessories which	▪ Fuses

may affect or obstruct the function of the seatbelt system ▪ Frayed or damaged webbing ▪ Missing or damaged button stop ▪ Seatbelt Pretensioners: retractors and lap belt (anchor point, driver only). For lap belt (anchor point) pretensioner webbing check, please refer to Lap belt (anchor point) pretensioner webbing check	▪ Wiring harness fault ▪ Correct engagement of electrical connectors ▪ Loose or corroded connections ▪ Warning lamp bulb(s) ▪ Impact sensor(s) ▪ Buckle sensor(s) ▪ Seatbelt Pretensioners: retractors and lap belt (anchor point, driver only) ▪ Belt tension sensor(s) ▪ Restraints control module
--	--

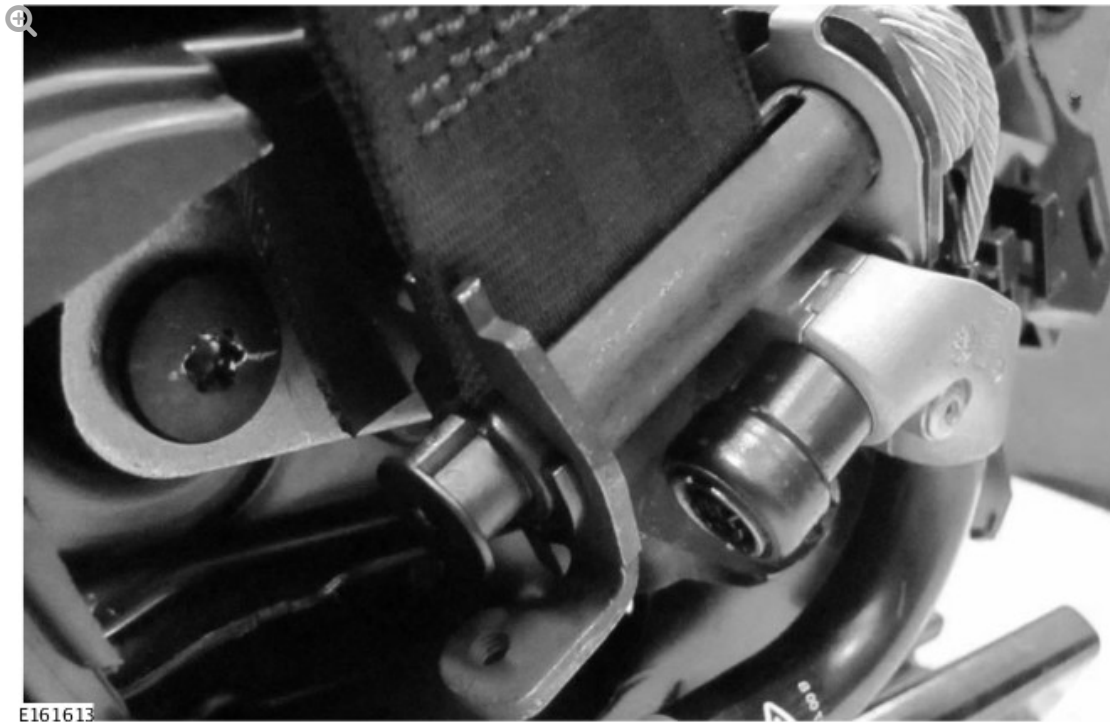
3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, carry out the test methods described below, alternatively check for diagnostic trouble codes and refer to the relevant diagnostic trouble code index

For a complete list of all diagnostic trouble codes that could be logged on this vehicle, please refer to section 100-00.

Lap belt (anchor point) pretensioner webbing check

Refer to the 501-20 removal and installation section of the workshop manual, remove any trim panels required to expose the lap belt (anchor point) pretensioner.



Be careful not to remove the webbing pin from the loop of the webbing. Make sure that the webbing pin is fitted correctly inside the loop of the webbing with the webbing pin and the loop slotted into the assembly.

SYMPTOM CHART FOR SEATBELT ROWS 1, 2 AND 3

SYMPTOM	POSSIBLE CAUSES	ACTION
Seatbelt jammed - Webbing tight	▪ Backlock effect-in action (webbing retracted quickly and came to sudden stop) ▪ Seatbelt retractor not installed correctly ▪ Rear centre seatbelt only. Mini-button (webbing travel limit stop) missing and seat squab has been moved causing tight fit ▪ Automatic locking retractor activated (clicking – during retraction only)	▪ GO to Pinpoint Test A. ▪ GO to Pinpoint Test F. ▪ GO to Pinpoint Test H. ▪ See the automatic locking retractor description below
Seat squab will not fold/jammed		▪ GO to Pinpoint Test H.

	NOTE: Rear centre seatbelt only Mini-button (webbing travel limit stop) missing and seat squab has been moved causing excessive tension	
Seatbelt jammed - Webbing loose	- Seatbelt webbing trapped in seat - Seatbelt retractor webbing guide loose - Twist in webbing - Interference in webbing routing - D-loop not rotating correctly	- GO to Pinpoint Test B. - GO to Pinpoint Test C. - GO to Pinpoint Test D. - GO to Pinpoint Test E. - GO to Pinpoint Test G.
Seatbelt - Intermittent jamming	Seatbelt retractor not installed correctly	GO to Pinpoint Test F.
Seatbelt - Slow retraction	- Seatbelt retractor webbing guide loose - Twist in seatbelt webbing - Interference in webbing routing - Seatbelt retractor not installed correctly - D-loop not rotating correctly - Foreign object/debris	- GO to Pinpoint Test C. - GO to Pinpoint Test D. - GO to Pinpoint Test E. - GO to Pinpoint Test F. - GO to Pinpoint Test G. - GO to Pinpoint Test E.
Seatbelt - Not retracting	- Seatbelt retractor webbing guide loose - Twist in seatbelt webbing - D-loop not rotating correctly - Interference in webbing routing - Foreign object/debris	- GO to Pinpoint Test C. - GO to Pinpoint Test D. - GO to Pinpoint Test G. - GO to Pinpoint Test E. - GO to Pinpoint Test E.
Seatbelt - Not extracting	- Backlock effect-in action (webbing retracted quickly and came to sudden stop) - Seatbelt retractor not installed correctly - Seatbelt retractor webbing guide loose - Twist in seatbelt webbing - D-loop not rotating correctly	- GO to Pinpoint Test A. - GO to Pinpoint Test F. - GO to Pinpoint Test C. - GO to Pinpoint Test D. - GO to Pinpoint Test G. - GO to Pinpoint Test E.

	▪ Interference in webbing routing ▪ Foreign object/debris ▪ Automatic locking retractor activated (clicking – during retraction only)	▪ GO to Pinpoint Test E. ▪ See the automatic locking retractor description below
Seatbelt - Noisy during operation	▪ Automatic locking retractor activated (clicking–during retraction only) ▪ Interference in webbing routing (rubbing)	▪ GO to Pinpoint Test B. ▪ GO to Pinpoint Test E.
Seatbelt buckle - Not latching / jammed	▪ Foreign object/debris	CAUTION: Do not insert any objects or tools into the buckle head ▪ GO to Pinpoint Test I.

INERTIA REEL SEATBELTS

The vehicle is equipped with (two row one), (three row two), and (two row three (seven seat versions only)) inertia reel seatbelts

These seatbelts are "**dual sensitive**" which means that they have:

- **Car sense system - A vehicle motion sensor, which locks the seatbelt webbing under braking, cornering, on steep hills and in adverse camber conditions, when parked on a steep incline or driveway or two wheels on a high curb**
- **Webb sense system - A webbing motion sensor, which locks when the seatbelt webbing is extracted suddenly**

The seatbelts in the following positions are equipped with an automatic locking retractor function:

CARLINE	MARKET	SEAT POSITION	AUTOMATIC LOCKING RETRACTOR INSTALLED	FROM MODEL YEAR
Defender (L316)	All	All	No	2007
Discovery / Range Rover Sport (L319/L320)	All	Driver	No	2008
Discovery / Range Rover Sport (L319/L320)	US	Passenger	Yes	2005
Discovery / Range Rover Sport (L319/L320)	All	Driver	No	2005
Discovery / Range Rover Sport (L319/L320)	ROW	Passenger	No	2005
Discovery (L319)	All	Row 2	Yes	2005
Discovery (L319)	All	Row 3	Yes	2005
Range Rover Sport (L320)	All	Row 2	Yes	2006
Freelander (L359)	All	Driver	No	2007
Freelander (L359)	ROW	Passenger	No	2007
Freelander (L359)	US	Passenger	Yes	2007
Freelander (L359)	ROW	Row 2	No	2007
Freelander (L359)	US	Row 2	Yes	2007
Range Rover Evoque (L538)	All	Driver	No	2011
Range Rover Evoque (L538)	ROW	Passenger	No	2011
Range Rover Evoque (L538)	US	Passenger	Yes	2011
Range Rover Evoque (L538)	ROW	Row 2	No	2011
Range Rover Evoque (L538)	US	Row 2	Yes	2011
Range Rover (L322)	All	Driver	No	2003

Range Rover (L322)	ROW	Passenger	No	2003
Range Rover (L322)	US	Passenger	Yes	2003
Range Rover (L322)	ROW	Row 2	No	2003
Range Rover (L322)	US	Row 2	Yes	2003
Range Rover Sport (L494)	All	Driver	No	2014
Range Rover Sport (L494)	ROW	Passenger	No	2014
Range Rover Sport (L494)	US	Passenger	Yes	2014
Range Rover Sport (L494)	ROW	Row 2	No	2014
Range Rover Sport (L494)	US	Row 2	Yes	2014
Range Rover Sport (L494)	ROW	Row 3	No	2014
Range Rover Sport (L494)	US	Row 3	Yes	2014
Range Rover (L405)	All	Driver	No	2013
Range Rover (L405)	ROW	Passenger	No	2013
Range Rover (L405)	US	Passenger	Yes	2013
Range Rover (L405)	ROW	Row 2	No	2013
Range Rover (L405)	US	Row 2	Yes	2013
Range Rover (L405)	All	Row 3	No	2013

The **automatic locking retractor function** is a feature to secure a child seat or heavy load to the seat

ACTIVATION	DEACTIVATION
	Automatic locking retractor is deactivated by allowing the webbing to retract until the clicking stops (close to park position)

NOTE: When automatic locking retractor is activated, no further webbing can be drawn from the seatbelt retractor, prior to disengagement of the automatic locking. This can be mistaken as a jammed seatbelt retractor	
Activated by total extraction of the webbing	
When activated the automatic locking retractor is identified by a clicking noise during webbing retraction	When deactivated the automatic locking retractor seatbelt changes state, from a static seatbelt to an automatic seatbelt

SEATBELT LOCKING TEST

With the vehicle stationary and on level ground take firm hold of the seatbelt webbing (on the tongue side of the upper seatbelt anchor) and withdraw sharply, **the retractor should lock**. Preventing further webbing release (**repeat this test 3 times**). Any seatbelt retractor which fails to lock **must not be used** and a **new seatbelt must be installed**.

DTC INDEX

For a list of diagnostic trouble codes that could be logged on this vehicle, please refer to Section 100-00 or for removal and installation/description and operation see Section 501-20

DIAGNOSTIC GUIDE INERTIA REEL SEATBELTS

PINPOINT TEST A : BACKLOCK	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	A1: BACKLOCK

	1 Visually inspect the condition of the suspect seatbelt
	2 Draw a maximum of 20mm of the webbing from the seatbelt retractor with moderate force. Then release the webbing
	3 Check for correct operation twice
	Does the webbing move freely then retract correctly? Yes No further action required No For first row seatbelt GO to Pinpoint Test C. For second and third row seatbelts GO to Pinpoint Test B.

PINPOINT TEST B : WEBBING - TRAPPED IN SEAT	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: WEBBING - TRAPPED IN SEAT	
	1 Visually inspect the condition of the suspect seatbelt
	2 Lift the seat base or release the seat backrest as required
	3 Free the trapped webbing, allow the webbing to retract Note: If the automatic locking retractor is activated, allow the webbing to retract until the clicking stops
	4 Check for correct operation twice
	Does the webbing move freely then retract correctly? Yes No further action required No GO to Pinpoint Test C.

PINPOINT TEST C : SEATBELT RETRACTOR - WEBBING GUIDE LOOSE	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: SEATBELT RETRACTOR - WEBBING GUIDE LOOSE	
	1 Refer to 501-20 removal and installation section of the workshop manual, remove any trim panels required to expose the D loop (anchor point) and seatbelt retractor
	2 Check the webbing is not trapped or twisted and is centrally located on the seatbelt retractor spindle
	3 Attempt to withdraw the webbing from the seatbelt retractor NOTE: If the seatbelt webbing is jammed, the automatic locking retractor

	could be engaged
	4 To release the automatic locking retractor, manually wind the webbing onto the spindle until the automatic locking retractor deactivates (clicking stops)
	5 Fully extract webbing
	6 Confirm webbing guide location is correct , Confirm the fixing lugs are correctly located in the retractor frame
	7 Allow webbing to retract
	8 Check for correct operation twice
	Does the webbing move freely then retract correctly? Yes Refer to the 501-20 removal and installation section of the workshop manual, reinstall any trim panels, ensure there are no obstructions and the webbing does not catch or rub. No further action required No GO to Pinpoint Test D.

PINPOINT TEST D : TWIST IN WEBBING	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: TWIST IN WEBBING	
	1 Refer to section 501-20 removal and installation section of the workshop manual, remove any trim panels required to expose the D loop (anchor point)
	2 Twist the webbing back the correct way in the loop
	3 Pass the twist through the pillar loop or escutcheon as required
	4 Check for correct operation twice
	Does the webbing move freely then retract correctly? Yes Refer to the 501-20 removal and installation section of the workshop manual, reinstall any trim panels, ensure there are no obstructions and the webbing does not catch or rub. No further action required No GO to Pinpoint Test E.

PINPOINT TEST E : INTERFERENCE - WEBBING ROUTING	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS

E1: INTERFERENCE - WEBBING ROUTING

	1 Refer to the 501-20 removal and installation section of the workshop manual, remove any trim panels required to expose the D loop (anchor point)
	2 Remove obstructions and foreign objects ensure the webbing does not catch or rub
	3 Confirm the seatbelt does not contact the wiring harness
	4 Check for correct operation twice
	Does the webbing move freely then retract correctly? Yes Refer to the 501-20 removal and installation section of the workshop manual, reinstall any trim panels, ensure there are no obstructions and the webbing does not catch or rub. No further action required No GO to Pinpoint Test F.

PINPOINT TEST F : SEATBELT RETRACTOR - INCORRECT INSTALLATION

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: SEATBELT RETRACTOR - INCORRECT INSTALLATION	
	1 Refer to the 501-20 removal and installation section of the workshop manual, remove any trim panels required to expose the D loop (anchor point) and the seatbelt retractor
	2 Refer to the 501-20 removal and installation section of the workshop manual, correctly reinstall the seatbelt retractor ensure that the locating "T bar" and "anti rotation pins" are correctly located
	3 Check for correct operation twice
	Does the webbing move freely then retract correctly? Yes Refer to the 501-20 removal and installation section of the workshop manual, reinstall any trim panels, ensure there are no obstructions and the webbing does not catch or rub. No further action required No Replace as required. Refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component

PINPOINT TEST G : D-LOOP NOT ROTATING CORRECTLY

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
-----------------	-------------------------

G1: D-LOOP NOT ROTATING CORRECTLY

	1 Refer to the 501-20 removal and installation section of the workshop manual, remove any trim panels required to expose the D loop (anchor point) and the seatbelt retractor
	2 Ensure there are no obstructions and the webbing does not catch or rub, the D loop (anchor point) rotates correctly and if installed the confirm the height adjuster operates correctly
	3 Check for correct operation twice
	Does the webbing move freely then retract correctly? Yes Refer to the 501-20 removal and installation section of the workshop manual, reinstall any trim panels, ensure there are no obstructions and the webbing does not catch or rub. No further action required No Replace as required. Refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component

NOTE:

This test applies to the rear centre seatbelt retractor installed in the **seat back**

PINPOINT TEST H : MINI BUTTON - MISSING/DAMAGED

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
H1: MINI BUTTON - MISSING/DAMAGED	
	1 Refer to the 501-20 removal and installation section of the workshop manual, remove the seat cushion and the plastic escutcheon at the top of the seat back (where the webbing exits to expose the lower anchor fixing point of the center seatbelt)
	2 Remove the lower anchorage of the seatbelt
	3 With the seat back correctly latched, allow up to 20mm webbing to retract, then extract the webbing
	Is the mini-button (webbing travel limit stop) correctly installed to the webbing and in good condition? Yes Feed the mini-button back through the plastic escutcheon if required. Correctly reinstall the escutcheon to the seat back, extract the webbing

then allow to retract, ensure the mini-button comes to rest outside the escutcheon stop

No

Replace as required. Refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component

CAUTION:

Do not insert any objects or tools into the buckle head

PINPOINT TEST I : SEATBELT BUCKLE - NOT LATCHING/JAMMED	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
I1: SEATBELT BUCKLE - NOT LATCHING/JAMMED	
	1 Visually inspect the buckle head for evidence of damage. If damaged replace as required
	2 Depress the buckle release (red button) and (Using a torch) carry out visual inspection for any evidence of debris/material or foreign objects in the buckle head
	3 If required remove the seatbelt buckle from the vehicle. Remove the seat. Remove the seatbelt buckle from the seat frame
	4 Do not insert any objects or tools into the buckle head With the buckle removed invert and attempt to shake out any debris
	5 Attempt to latch the tongue in the buckle
	Does the seatbelt buckle operate correctly Yes Reinstall any components, no further action required No Replace the seatbelt buckle, Refer to section 501 20